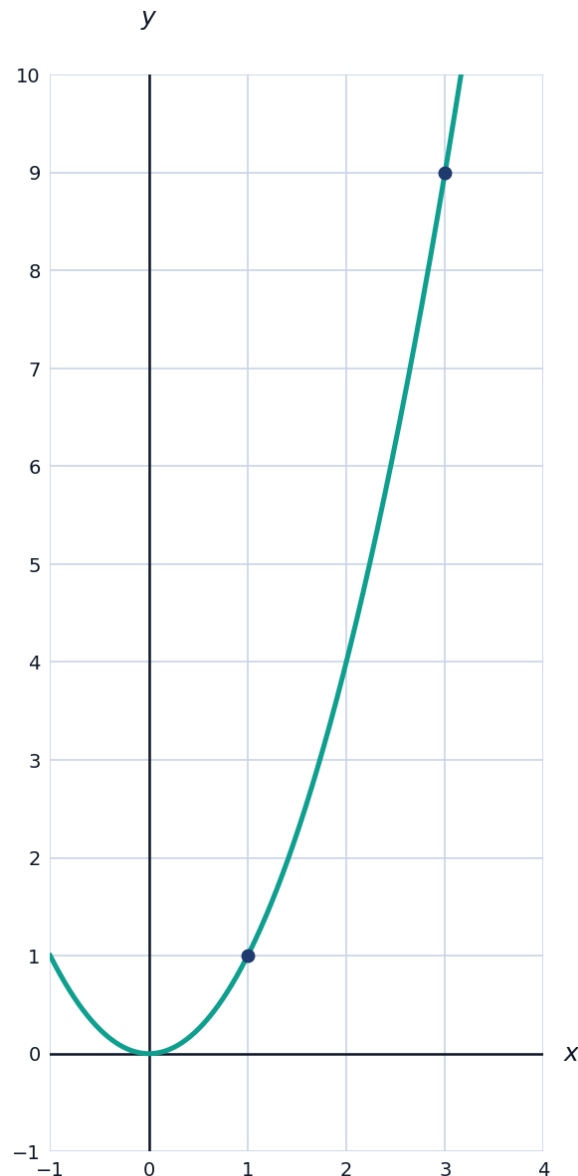


# The Derivative from First Principles



## The definition of the derivative

- 1 Use the limit definition  $f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$  to find  $f'(x)$  for  $f(x) = x^2 + 3x$ .
- 2 Use the limit definition to find  $f'(x)$  for  $f(x) = \sqrt{x}$  (for  $x > 0$ ).
- 3 The graph shows  $f(x) = x^2$  with the secant line through  $(1, 1)$  and  $(3, 9)$ .



- a Find the slope of the secant line.
- b Find  $f'(2)$  using the derivative, and compare it to the secant slope.

## Differentiability

- 4 Explain why  $f(x) = |x|$  is not differentiable at  $x = 0$ , using one-sided derivatives.
- 5 State the relationship between differentiability and continuity: if a function is differentiable at a point, must it be continuous there? Is the converse true?